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Modular crate components and systems

Abstract

Modular crate components and systems can include panels, connectors, and fasteners useable to create crates of different sizes. The panels can include a plurality of teeth along each of the edges of the panel that can interlock or intermesh with a plurality of teeth of an adjacent panel. At least some of the teeth can include posts that are configured to engage with the connectors to facilitate coupling the panels together. The connectors can include an arrangement of slots that are positioned and configured to engage with the posts included on at least some of the teeth. Each connector can also be configured with an opening configured to receive a fastener therethrough.

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Background/Summary

PRIORITY APPLICATION (1) This application claims priority to U.S. Provisional Application No. 63/655,489, filed Jun. 3, 2024. Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57.

BACKGROUND

Field

(1) This application relates to modular crate components and systems that can be used to construct crates, for example, for shipping or storage of items, of various sizes.

Description

(2) Crates are often used to protect items, for example, during shipping or storage. Typically, crates are constructed from wood and the size of the crate is custom made to accommodate the item to be placed therein.

SUMMARY

(3) This application describes modular crate components and systems that can be used to quickly and efficiently construct crates of different sizes from a set of modular components. The modular components can include panels, connectors, and fasteners (such as bolts). By using different numbers and arrangements of the panels, connectors, and fasteners, crates of different sizes can be constructed.

(4) The panels can include a plurality of teeth along each of the edges of the panel that can interlock or intermesh with a plurality of teeth of an adjacent panel. At least some of the teeth can include posts that are configured to engage with the connectors to facilitate coupling the panels together. Adjacent panels can be arranged at either a 180-degree configuration (allowing for construction of flat sides of different lengths and widths) or 90-degree configurations (allowing for the coupling of the different sides of the crates and the formation of corners).

(5) The connectors can include an arrangement of slots (e.g., T-slots) that are positioned and configured to engage with the posts included on at least some of the teeth. That is, in some

embodiments, the connectors can include slots configured to engage with and retain the posts to connect and orient panels relative to each other. Each connector can also be configured with an opening configured to receive a fastener (such as, for example, an M8 bolt). The connectors can be advantageously configured such that the connector can be engaged with two, three, or four adjacent panels (for example, through engagement of the posts and slots) while only requiring a single fastener to secure the connector in place. For example, the slots of the connector can be engaged with posts of multiple panels, while the connector is secured with the fastener to only a single panel. In this way, construction of crates from these modular components is simplified as the total number of fasteners that need to be used is greatly reduced.

(6) These and other aspects of the disclosure will become apparent from the following detailed description, drawings, and appended claims. This summary is provided for illustrative and introductory purposes and is not intended to be limiting.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The features and advantages of the modular crate components and systems described herein will become more fully apparent from the following description and appended claims, taken in conjunction with the accompanying drawings. These drawings depict only several embodiments in accordance with the disclosure and are not to be considered limiting of its scope. In the drawings, similar reference numbers or symbols typically identify similar components, unless context dictates otherwise. In some instances, the drawings may not be drawn to scale.

(2) FIG. 1 is a perspective view of an inner surface of an embodiment of a panel for a modular crate system.

(3) FIG. 2 is a perspective view of an outer surface of an embodiment of the panel for the modular crate system of FIG. 1.

(4) FIG. 3 is a detail perspective view of an inner surface of a corner of the panel of the modular crate system of FIG. 1, illustrating features of a plurality of teeth of the panel in greater detail.

(5) FIG. 4 is a top perspective view of an embodiment of a connector for a modular crate system.

(6) FIG. 5 is a bottom perspective view of the connector of FIG. 4.

(7) FIG. 6 is a perspective view of a fastener for use with a modular crate system.

(8) FIG. 7 is a perspective view showing a first example of a modular crate assembled using a plurality of the panels of FIG. 1, a plurality of connectors of FIG. 4, and a plurality of fasteners of FIG. 6.

(9) FIG. 8 is a perspective view showing a second example of a modular crate assembled using a plurality of the panels of FIG. 1, a plurality of connectors of FIG. 4, and a plurality of fasteners of FIG. 6.

(10) FIG. 9 is a perspective view showing a third example of a modular crate assembled using a plurality of the panels of FIG. 1, a plurality of connectors of FIG. 4, and a plurality of fasteners of FIG. 6.

(11) FIG. 10 is a detail view showing an enlarged portion of a corner of a modular crate, such as the modular crate of FIGS. 7-9.

(12) FIG. 11 is an exploded detail view of the corner of FIG. 10.

(13) FIG. 12 is a detail view showing an enlarged portion of an edge of a modular crate, such as the modular crate of FIGS. 8-9.

(14) FIG. 13 is an exploded detail view of the edge of FIG. 12.

(15) FIG. 14 is a detail view showing an enlarged portion of an outer side of a flat joint between four adjacent panels of a modular crate, such as the modular crate of FIG. 9.

(16) FIG. 15 is a detail view showing an enlarged portion of an inner side of the flat joint of FIG.

14.

(17) FIG. 16 shows an example modular crate with example handles and wheels attached to outer surfaces thereof.

(18) FIG. 17 shows an example modular crate with example handles and legs attached to outer surfaces thereof.

(19) FIG. 18 shows an example modular crate with example handles and a pallet base attached to outer surfaces thereof.

(20) FIG. 19 illustrates an example joint between two modular panels that includes a rod to strengthen the joint.

(21) FIG. 20 is an exploded view of the joint of FIG. 19.

DETAILED DESCRIPTION

(22) This application describes modular crate components and systems that can be used to quickly and efficiently construct crates from a set of modular components. As desired, a user can construct crates with different sizes and configurations by selecting different numbers and arrangements of the modular components.

(23) The modular crate systems described herein provide a number of advantages over the traditional methods for constructing crates. For example, traditional crates are generally made from wood and require specific tooling and woodworking experience to manufacture. Thus, wooden crates are heavy, expensive, and labor intensive to produce. Further, wooden crates are typically either available in standard, non-customizable sizes or custom fabricated in specific sizes on an as-needed basis. This means that wooden crates either are only likely to accommodate standard sized items or must be custom fabricated for uniquely sized items. Additionally, wooden crates are often single use items which may be destroyed as they are opened. The modular crate systems described herein advantageously use a small set of lightweight modular components which can be assembled quickly in a variety of configurations to produce crates of different sizes, without requiring significant tooling, time, or expertise. Further, the modular crate components can be reusable, further reducing waste and cost. In some embodiments, at least some of the modular crate components, such as the modular panels, for example, are made from plastic. Use of plastic can be advantageous because it is more resistant to fire, water, and insects than traditional wooden crates.

(24) The modular components can include panels, connectors, and fasteners (such as bolts).

(25) In some embodiments, a panel for a modular crate system can include a rectangular body. The rectangular body can include an inner surface and an outer surface. The rectangular body can also include a first edge, a second edge, a third edge, and a fourth edge. In some embodiments, the rectangular body can be square. In some embodiments, the square can be 40 inches by 40 inches by 1.5 inches, although other dimensional sizes are possible.

(26) Each edge of the panel can include a plurality of teeth extending outwardly from the edge. The plurality of teeth can be spaced apart such that the plurality of teeth is configured to intermesh with a plurality of teeth of an adjacent panel of the modular crate system to form at least a portion of a modular crate. Adjacent panels can be positioned at either a 180-degree configuration relative to each other in order to form a flat side or a 90-degree configuration relative to each other in order to form a corner. The plurality of teeth along each edge can also include a tooth positioned at one end of the edge that comprises a post configured to engage with a slot of a connector to couple the panel to the adjacent panel.

(27) In some embodiments, the plurality of teeth positioned along each edge of the panel include at least two different types of teeth. For example, the plurality of teeth can include one or more teeth of a first tooth type and one or more teeth of a second tooth type. In some instances, at least some of the plurality of teeth alternate between the first tooth type and the second tooth type along the edge of the panel. In some embodiments, teeth of the first tooth type include flanges that extend distally from a distal face of the tooth along edges aligned with the inner and outer surfaces of the panel. In some embodiments, teeth of the second tooth type also include flanges that extend distally

from a distal face of the tooth along edges aligned with the inner and outer surfaces of the panel, but also include flanges that extend orthogonally from inner and outer faces of the teeth at the distal end of the tooth.

(28) In some embodiments, when two panels are arranged adjacently in a 180-degree (flat) configuration, the distally extending flanges of the first and second teeth types overlap a portion or are received within corresponding recesses on the inner and outer surfaces of the adjacent panel. This overlap facilitates and strengthens a flat connection between the adjacent panels. In some embodiments, when two panels are arranged adjacently in a 90-degree (corner) configuration, the orthogonally extending flanges of the second teeth type overlap a portion or are received within corresponding recesses on the inner and outer surfaces of the adjacent panel. This overlap facilitates and strengthens a corner connection between the adjacent panels.

(29) The plurality of teeth can also include one or more teeth that are configured for further engagement with a connector of the modular crate system. For example, a tooth on one end of each edge of the panel can include a post that extends from a side surface of the tooth. The post can be configured to engage with a slot of the connector. The post can take many forms. In some embodiments, the post comprises a cylindrical projection terminating in a circular flange. In some embodiments, the post further comprises an opening extending along a longitudinal axis of the post. The opening is threaded and configured to receive a fastener (e.g., a bolt). In some embodiments, the bolt comprises an M8 bolt, although other bolts or fasteners can be used. The fastener can thus be used to secure the connector to the post of a panel.

(30) In some embodiments, a connector for a modular crate system can include a cubic body comprising an upper surface and an opposed lower surface, and four side surfaces extending between the upper surface and the lower surface.

(31) The connector can further include an opening extending through the cubic body from the upper surface to the lower surface. The opening can be configured to receive a shaft of a bolt therethrough. In some embodiments, the opening is not threaded, although in other embodiments, the opening can be threaded. In some embodiments, the bolt comprises an M8 bolt, although other bolts or fasteners can be used. In some embodiments the upper surface comprises a recess configured to receive a head of a bolt inserted through the opening. As mentioned above, an advantage of the modular crate system is that only a single fastener is required for each connector. That is, one fastener is used to secure the connector to one panel, but the connector is further engaged with one, two, three, or four other panels through the engagement of its slots with corresponding posts of the other panels.

(32) Accordingly, the connector can further include, for each of the side surfaces, a slot formed into the side surface and extending from the lower surface towards the upper surface. In some embodiments, the slot comprises a T-shaped slot. In some embodiments, The T-shaped slot is configured such that the circular flange at the end of the post can be retained therein.

(33) In some embodiments, the cubic body of the connector is 1.5 inches by 1.5 inches by 1.5 inches. Other dimensions for the connector are possible. In some embodiments, the dimensions of the connector are configured to correspond with the thickness of the panels.

(34) The following discussion presents detailed descriptions of the several embodiments of a modular crate components and systems as shown in the figures. These embodiments are not intended to be limiting, and modifications, variations, combinations, etc., are possible and within the scope of this disclosure.

(35) FIGS. 1-3 illustrate various views of an embodiment of a panel **100** of a modular crate system. FIG. 1 is a perspective view of an inner surface **102** of the panel **100**, FIG. 2 is a perspective view of an outer surface **104** of the panel **100**, and FIG. 3 is a detail view of a corner of the panel **100**.

(36) Beginning with FIGS. 1 and 2, showing the inner surface **102** and the outer surface **104** of the panel **100**, respectively, in the illustrated embodiment, the panel **100** generally comprises a rectangular body having a thickness measured between the outermost portions of the inner surface

102 and the outermost portions of the outer surface **104** and bounded on its lateral edges by a first edge **106a**, a second edge **106b**, a third edge **106c**, and a fourth edge **106d**. Because, in some embodiments, each of the first edge **106a**, the second edge **106b**, the third edge **106c**, and the fourth edge **106d** are substantially similar, reference numeral **106** is used herein to refer to any edge generically. Further, while the inner surface **102** and the outer surface **104** are referred to herein as surfaces, this term is used broadly. For example, as shown, not all features on the inner surface **102** (FIG. **1**) and the outer surface **104** (FIG. **2**) need lie in a single plane. In the illustrated embodiment, the rectangular body comprises a square shape, although this need not be the case in all embodiments. Further, various dimensions for the panel **100** are possible. For example, in an embodiment, the panel **100** is 40 inches by 40 inches by 1.5 inches, although this is provided only by way of example and other sizes are possible.

(37) As shown in FIGS. **1** and **2**, each edge **106** of the panel **100** is provided with a plurality of teeth **108**. In the illustrated embodiment, the panel **100** includes two types of teeth: a first tooth type **108a** and a second tooth type **108b**. Reference numeral **108** is used to refer to any of the teeth generically, while reference numerals **108a** and **108b** are used to refer to a tooth of the first or second tooth type, respectively. The teeth **108** will be described in more specific detail below with reference to the detail view of FIG. **3**, which shows an enlarged corner of the panel **100**. FIGS. **1** and **2**, however, show that the teeth **108** can be spaced apart along the edges **106** of the panel **100**. In general, the teeth **108** are spaced apart such that they are configured to intermesh with teeth of an adjacent panel **100** of the modular crate system when the panels **100** are assembled. Examples of assembled crates are shown in FIGS. **7-9** and example joints between adjacent panels **100** are shown in more detail in FIGS. **10-15**, which are described in more detail below. To accommodate intermeshing of teeth **108** of adjacent panels, the width between teeth **108** on a panel **100** can generally be equal to the width of each tooth **108**. In this manner, when intermeshed, teeth **108** generally do not comprise gaps therebetween.

(38) As also shown in FIGS. **1** and **2**, the panel **100** can include at least one tooth **108** on each edge **106** that includes a post **110**. The post **110** can be configured for engagement with a connector **200** as shown in FIGS. **4** and **5**. The post **110** can also be configured to receive a fastener, for example, the bolt **300**, shown in FIG. **6**. The post **110**, connector **200**, and fastener **300** will be described in more detail further below. In the illustrated embodiment of the panel **100** of FIGS. **1** and **2**, the tooth **108** that includes the post **110** is the tooth positioned at one end of the edge **106**. In the illustrated embodiment, the post **110** extends from a side surface of the tooth **108**.

(39) The panel **100** can be configured to be lightweight, yet rigid and durable. In the illustrated embodiment, the panel **100** comprises a frame or boarder **114** that extends around the edges **106** of the panel **100**. As shown in FIG. **1**, the boarder **114** can comprise a smooth surface lying in a plane on the inner surface **102** of the panel **100**. As shown in FIG. **2**, the boarder **114** can comprise a plurality of recessed squares on the outer surface **104** of the panel **100**. This configuration provides a light weight, while maintaining structural rigidity. In the illustrated embodiment, the panel **100** also comprises interior crossbeams **116** that extend across the panel **100** connecting opposite sides of the boarder **114**. As shown in FIG. **1**, the crossbeams **116** can comprise a smooth surface lying in a plane on the inner surface **102** of the panel **100**. As shown in FIG. **2**, the crossbeams **116** can comprise a plurality of support baffles on the outer surface **104** of the panel **100**. This configuration provides further provides a light weight, while maintaining structural rigidity. In some embodiments, the panels **100** may be made from a plastic material.

(40) As shown in FIGS. **1** and **2**, the boarder **114** and crossbeams **116** divide the panel into four interior sections **118**. As shown in FIG. **1**, the interior sections **118** can comprise a plurality of recessed squares on the inner surface **102** of the panel **100**. As shown in FIG. **2**, the interior sections **118** can comprise a smooth surface lying in a plane on the outer surface **104** of the panel **100**. This configuration again provides a light weight, while maintaining structural rigidity. In some embodiments, the flat surfaces of the crossbeams **116** and/or the boarder **114** on the inner surface

102 can provide surfaces to which adhesives can be adhered, for example, to apply foam blocks or other cushioning or supporting materials. The recessed portions of the crossbeams **116** and/or boarder **114** on the outer surface **104** can be used as grab handles to facilitate handling of a modular crate.

(41) FIG. 2 also illustrates that the outer surface **104** of the panel **100** can include a plurality of attachment locations **120** that can be used to attach components to the exterior of a crate formed by the panel **100**. In the illustrated embodiment, the attachment locations **120** comprise threaded holes configured to receive fasteners, such as bolts. The fasteners can be used to secure various items such as wheels, pallet legs, handles, feet, etc., to the outer surface **104** of the panel **100**. Examples are shown in FIGS. 16-18 which are described below. In the illustrated embodiment, the attachment locations **120** are provided at the corners of the boarder **114** and at the center of the crossbeams **116**, although other numbers and locations for the attachment locations **120** can also be used. In some embodiments, attachment locations **120** can be included on the inner surface **102** of the panel **100**. In some embodiments, the inner surface **102** of the panel **100** can be configured for use with internally attaching components. For example, in some embodiments, the internally attaching components can be configured to engage with the square recesses in the interior sections **118**. For example, crossbeams or supports that are used to support an item within the crate or divide the interior of the crate into smaller sub compartments can be configured to engage with the square recesses in the interior sections **118**.

(42) In some embodiments, for example as illustrated, the panel **100** can include recesses **112**, such as the recesses **112a** of FIG. 1 and recesses **112b** of FIG. 2 on the boarder **114** at locations positioned between the teeth **108**. These recesses **112** can be configured to receive features of the teeth **108** as will be described below with reference to FIGS. 3 and 10-15.

(43) In some embodiments, the panel **100** can be made from a rigid plastic material, although other materials are also possible.

(44) FIG. 3 is a detail perspective view of the inner surface **102** of a corner of the panel **100** of FIG. 100. FIG. 3 will now be described with a particular emphasis on features of the teeth **108** (e.g., of the first tooth type **108a** and the second tooth type **108b**) and the post **110**.

(45) In the enlarged view of FIG. 3, the features of the first tooth type **108a** are more visible. As shown, in the illustrated embodiment, teeth of the first tooth type **108a** can include distally extending flanges **122** that extend from a distal face of the tooth **108a**. A first of the distally extending flanges **122** can be positioned on an edge of the distal face of the tooth on the inner side of the panel **100**, and a second of the distally extending flanges **122** can be positioned on an edge of the distal face of the tooth on the outer side of the panel **100**. When two panels **100** are joined in a flat, for example, 180-degree configuration, the distally extending flanges **122** overlap with or are at least partially received in the recesses **112a**, **112b** on the inner surface **102** and outer surface **104** of the panel **100**, respectively. This overlap or engagement strengthens the joint between adjacent panels **100**.

(46) FIG. 3 also illustrates that, in some embodiments, the first tooth type **108a** can include an opening **124** that extends through the tooth **108a** from the inner side to the outer side. In the illustrated embodiment, the opening is square, although this need not be the case in all embodiments. In some embodiments, the opening **124** can be omitted or oriented differently. In some embodiments, the teeth **108a** are configured with the opening **124** such that they are provided with some flexibility. The flexibility can allow the teeth **108** to interlock together and unlock more easily.

(47) Turning now to the features of the second tooth type **108b**, teeth of the second tooth type **108b** can include distally extending flanges **126**, similar to the distally extending flanges **122** of the first tooth type **108a** previously described. For example, the second tooth type **108b** can include distally extending flanges **126** that extend from a distal face of the tooth **108b**. A first of the distally extending flanges **126** can be positioned on an edge of the distal face of the tooth on the inner side

of the panel **100**, and a second of the distally extending flanges **126** can be positioned on an edge of the distal face of the tooth on the outer side of the panel **100**. When two panels **100** are joined in a flat, for example, a 180-degree configuration, the distally extending flanges **126** overlap with or are at least partially received in the recesses **112a**, **112b** on the inner surface **102** and outer surface **104** of the panel **100**, respectively. This overlap or engagement strengthens the joint between adjacent panels **100**.

(48) As shown in FIG. 3, the second tooth type **108b** also includes orthogonally extending flanges **128**. A first of the orthogonally extending flanges **128** extends orthogonally from a distal end of the inner face of the tooth **108b**, and a second of the orthogonally extending flanges extends orthogonally from the distal end of an outer face of the tooth **108b**. When two panels **100** are joined in a corner configuration, for example, a 90-degree configuration, the orthogonally extending flanges **128** overlap with or are at least partially received in the recesses **112a**, **112b** on the inner surface **102** and outer surface **104** of the panel **100**, respectively. This overlap or engagement strengthens the joint between adjacent panels **100**. In some embodiments, for example, as illustrated, the distally extending flanges **126** and the orthogonally extending flanges **128** can be connected and supported with support buttresses **130** as shown.

(49) FIG. 3 also illustrates that, in some embodiments, the second tooth type **108b** can include a longitudinal opening **132** that extends through the tooth **108b** from the one side to the other in a longitudinal direction that is parallel with the edge **106** of the panel. In the illustrated embodiment, the longitudinal opening **132** is square, although this need not be the case in all embodiments. In some embodiments, the teeth **108b** are configured with the longitudinal opening **132** such that they are provided with some flexibility. The flexibility can allow the teeth **108** to interlock together and unlock more easily. In some embodiments, the opening **132** can be omitted or oriented differently. In the illustrated embodiment, the second tooth type **108b** also includes transverse slits **134** positioned on the inner and outer faces of the tooth **108b**. Such transverse slits **134** may be included to reduce the cost and complexity of an injection molding process that is used to form the panel **100**. These need not be present in all embodiments.

(50) As shown in FIG. 3, in the illustrated embodiment, the post **110** can include a projection **136**, which can be cylindrical or other shapes, extending from a side surface of a first tooth **108** on the edge **106** of the panel. The cylindrical projection **136** can terminate at a flange **138**, which can be circular or other shapes, having a wider diameter than that of the cylindrical projection **136**. The post **110** is configured to be retained within a slot of the connector, such as the connector shown in FIGS. 4 and 5. Although the post **110** is shown as a cylindrical projection and circular flange, other shapes for the post **110** are possible.

(51) In the illustrated embodiment, an opening **140** extends through the post **110** along a longitudinal axis of the post **110**. The opening **140** can be configured to receive a fastener, such as the bolt **300** of FIG. 6. For example, in some embodiments, an internal surface of the opening **140** can be threaded for engagement with a threaded shaft of the bolt **300**. In some embodiments, the opening **140** can extend entirely through the tooth **108** and all of the teeth along the edge. In such embodiments, a rod can be inserted through the opening and through the intermeshed teeth **108** of two adjacent panels **100** to further strengthen a connection between the panels **100**, for example, as shown in FIGS. 19 and 20 described below.

(52) FIGS. 4 and 5 show top perspective and bottom perspective views, respectively, of an embodiment of a connector **200** for a modular crate system. The connector **200** can be used to facilitate securement and connection between two or more of the panels **100**. Advantageously, in some embodiments, only a single fastener or bolt is needed to secure the panels **100** and the connector **200**.

(53) As shown in FIGS. 4 and 5, the connector **200** can comprise a cubic body having an upper surface **202**, a lower surface **204**, and four side surfaces **206** extending between the upper and lower surfaces **202**, **204**. As shown, each of the side surfaces **206** can include a slot **208** formed

therein. Further, an opening **212** extends through the connector **200** from the upper surface **202** to the lower surface **204**.

(54) The cubic body of the connector **200** can be any size that corresponds with the size of the panels **100**. For example, typically, the length, width, and height dimensions of the connector **200** would correspond with the thickness of the panel **100**. For example, where the panel **100** is 40 inches by 40 inches by 1.5 inches, the connector **200** can be 1.5 inches by 1.5 inches by 1.5 inches. In some embodiments, the connector **200** can be the same size as the teeth **108**.

(55) As shown, the opening **212** extends through the connector **200** from the upper surface **202** to the lower surface **204**. The opening **212** can be configured to receive a fastener, such as the bolt **300** of FIG. 6, therethrough. In use, the bolt **300** is inserted through the opening **212** of the connector **200**. The bolt **300** is selected such that a portion of the bolt **300** extends proud from the lower surface **204** of the of the connector **200**. The proud portion of the bolt **300** can be engaged with the opening **140** of the post **110**. In some embodiments, the opening **212** through the connector **200** is not threaded. As shown in FIG. 4, in some embodiments, the upper surface **202** of the connector **200** includes a recessed portion **214** such that the head of the bolt **300** can be countersunk below the upper surface **202**. As further illustrated in FIG. 5, the lower surface **204** of the connector **200** can include a counterbore **216** that is configured to receive the post **110** therein, such that when the connector **200** is secured to the panel **100**, the lower surface **204** of the connector can be flush with the side surface of the tooth **108** from which the post **110** extends.

(56) Each of the side surfaces **206** of the connector **200** can include a similar slot **208**. The slot can extend from the lower surface **204** and up through the side surface **206** towards the upper surface **202**. That is, the lower portion of the slot **208** is open on the lower surface **204** and the upper portion of the slot **208** is closed on the upper surface **202**. In the illustrated embodiment, the slot comprises a T-slot. The T-slot is configured to receive and retain the circular flange **138** within the slot **208**. In the illustrated embodiment, an upper portion of the slot **208** is rounded. Other shapes and configurations for the slot **208** are possible that correspond and engage with the post **110**.

(57) FIG. 6 is a perspective view of a fastener or bolt **300** for use with a modular crate system. As shown, the bolt **300** can include a head **302** and threaded shaft **304**. In some embodiments, the bolt **300** comprises an M8 bolt, although other sizes and types of bolts and fasteners can be used.

(58) FIG. 7 is a perspective view showing a first example of a modular crate assembled using a plurality of the panels **100** of FIG. 1, a plurality of connectors **200** of FIG. 4, and a plurality of fasteners **300** of FIG. 6. In this example, six panels **100** are assembled into a cube. Each panel is joined with its adjacent panels **100** at 90-degree angles. A connector **200** and fastener **300** are included each corner of the crate such that six connectors **200** and six fasteners **300** are used. Each connector **200** is engaged with three panels **100** to form a corner as shown in FIGS. 10 and 11, which are described below. Here, if the panels are 40 inches by 40 inches by 1.5 inches, the resulting crate is 40 inches by 40 inches by 40 inches. Of course, these dimensions are provided only by way of example.

(59) FIG. 8 is a perspective view showing a second example of a modular crate assembled using a plurality of the panels **100** of FIG. 1, a plurality of connectors **200** of FIG. 4, and a plurality of fasteners **300** of FIG. 6. In this example, ten panels **100** are coupled by twelve connectors **200** and fasteners **300**. At each corner of the crate, three panels **100** are joined at 90-degrees to form a corner, for example, as shown in FIGS. 10 and 11. At the midline of each long edge of the crate, four panels are joined with a joint as shown in FIGS. 12 and 13. Here, if the panels are 40 inches by 40 inches by 1.5 inches, the resulting crate is 80 inches by 40 inches by 40 inches. Of course, these dimensions are provided only by way of example.

(60) FIG. 9 is a perspective view showing a third example of a modular crate assembled using a plurality of the panels **100** of FIG. 1, a plurality of connectors **200** of FIG. 4, and a plurality of fasteners **300** of FIG. 6. In this example, sixteen panels **100** are joined by eighteen connectors **200** and fasteners **300**. At each corner of the crate, three panels **100** are joined at 90-degrees to form a

corner, for example, as shown in FIGS. **10** and **11**. At the midline of each long edge of the crate, four panels **100** are joined with a joint as shown in FIGS. **12** and **13**. At the center of each large side of the crate, four panels **100** are joined with a joint as shown in FIGS. **14** and **15**. Here, if the panels are 40 inches by 40 inches by 1.5 inches, the resulting crate is 80 inches by 80 inches by 40 inches. Of course, these dimensions are provided only by way of example.

(61) The examples of crates provided in FIGS. **7-9** illustrate only some of the types and sizes of crates that can be assembled using the modular components as described herein. The modular components can be assembled in various configurations as desired to create custom sized crates as needed.

(62) FIG. **10** is a detail view showing an enlarged portion of a corner of a modular crate, such as the modular crate of FIGS. **7-9**. FIG. **11** is an exploded detail view of the corner of FIG. **10**. In this example, to form a corner, three panels **100** are arranged, each at 90-degrees with respect to the other two. A connector **200** is used at the intersection of the three panels **100** and is secured by a single fastener **300**. With reference to the exploded view of FIG. **11** and the orientation shown in the figure, the connector **200** is engaged with the upper panel **100** and the right panel **100** via engagement of the slots **208** (on the rear of the connector **200** relative to the orientation shown) with the posts **110** of the panels **100**. The post **110** of the left panel **100** is received within the counterbore **216** on the bottom surface **204** of the connector **200** and the fastener **300** is inserted through the opening **212** of the connector and engaged with the opening **140** of the post **110** of the left panel **100**. As shown, advantageously the connector **200** retains all three panels **100** while only requiring one fastener **300**.

(63) FIG. **12** is a detail view showing an enlarged portion of an edge of a modular crate, such as the modular crate of FIGS. **8-9**. FIG. **13** is an exploded detail view of the edge of FIG. **12**. In this example, to form an edge, four panels **100** are arranged as shown in FIG. **12**. That is, each panel is arranged at 180-degrees relative to one adjacent panel and 90-degrees relative to the other adjacent panel. A connector **200** is used at the intersection of the four panels **100** and is secured by a single fastener **300**. With reference to the exploded view of FIG. **13** and the orientation shown in the figure, the connector **200** is engaged with the upper right panel **100** and the lower right and left panels **100** via engagement of the slots **208** the posts **110** of these panels **100**. The post **110** of the upper left panel **100** is received within the counterbore **216** on the bottom surface **204** of the connector **200** and the fastener **300** is inserted through the opening **212** of the connector and engaged with the opening **140** of the post **110** of the upper left panel **100**. As shown, advantageously the connector **200** retains all four panels **100** while only requiring one fastener **300**.

(64) FIG. **14** is a detail view showing an enlarged portion of a flat joint between four adjacent panels of a modular crate, such as the modular crate of FIG. **9**. FIG. **15** is a detail view showing an enlarged portion of an inner side of the flat joint of FIG. **14** . . . In this example, four panels are arranged at 180-degrees relative to each other. A connector **200** is used at the intersection of the four panels **100**. The connector **200** is engaged with the posts **110** of each panel **100** via the slots **208**. A fastener **300** is inserted through the opening **212** of the connector and engaged with a plate **250** on the opposite side of the panels **100**. The plate **250** includes arms **252** that insert into the openings **124** of the teeth **108** to secure the panels **100** together.

(65) FIG. **16** shows an example modular crate with example handles **402** and wheels **404** attached to outer surfaces thereof. The handles **402** and/or wheels **404** can be selectively attachable to the outer surfaces **104** of the crate via the attachment locations **120** described above with reference to FIG. **2**. For example, the handles **402** and/or wheels **404** can include threaded bolts that are configured to be received in the attachment locations **120**. The handles **402** and wheels **404** shown in FIG. **16** are provided by way of example and may take other forms than those illustrated. FIG. **17** shows another example modular crate with example handles **402** and legs **406** attached to outer surfaces thereof. Again, the handles **402** and/or legs **406** can be configured to attach to the attachment locations **120**. The handles **402** and legs **406** can take other forms than those illustrated.

For example, the legs **406** can be longer or shorter or wider or narrower than illustrated. Further, the legs **406** can be hard or soft legs. FIG. **18** shows another example modular crate with example handles **402** and a pallet base **408** attached to outer surfaces thereof. The handles **402** and/or the pallet base **408** can be configured to attach to the attachment locations **120**. The pallet base **408** can be configured for use with a forklift or pallet jack to further facilitate handling of the crate. FIGS. **16-18** illustrate various external devices that can be attached to the exterior of a modular crate, other devices are also possible. Further, the external devices can quickly and easily be installed, swapped, and/or removed to further allow customization of the crates.

(66) FIG. **19** illustrates an example joint between two modular panels **100** that includes a rod **502** to strengthen the joint. FIG. **20** is an exploded view of the joint of FIG. **19**. As shown in FIGS. **19** and **20**, a rod **502** can be inserted through intermeshed teeth **108** of two adjacent panels **100**. The rod **502** can comprise, for example, a metal rod, such as a steel rod, although other materials can also be used. Because the rod extends through the intermeshed teeth **108**, for example, through openings in each tooth **108**, the rod **502** secures the two panels **100** together. The rod **502** can be inserted through the opening in the post **110**. The rod **502** can be retained in place by a connector **200** (not shown in FIGS. **19** and **20**) when the connector **200** is installed on the post **110**.

(67) The foregoing description details certain embodiments of the systems, devices, and methods disclosed herein. It will be appreciated, however, that no matter how detailed the foregoing appears in text, the systems, devices, and methods can be practiced in many ways. As is also stated above, it should be noted that the use of particular terminology when describing certain features or aspects of the invention should not be taken to imply that the terminology is being re-defined herein to be restricted to including any specific characteristics of the features or aspects of the technology with which that terminology is associated.

(68) It will be appreciated by those skilled in the art that various modifications and changes may be made without departing from the scope of the described technology. Such modifications and changes are intended to fall within the scope of the embodiments. It will also be appreciated by those of skill in the art that parts included in one embodiment are interchangeable with other embodiments; one or more parts from a depicted embodiment can be included with other depicted embodiments in any combination. For example, any of the various components described herein and/or depicted in the figures may be combined, interchanged or excluded from other embodiments.

(69) The above description discloses several methods and materials of the present invention. This invention is susceptible to modifications in the methods and materials, as well as alterations in the fabrication methods and equipment. Such modifications will become apparent to those skilled in the art from a consideration of this disclosure or practice of the invention disclosed herein. Consequently, it is not intended that this invention be limited to the specific embodiments disclosed herein, but that it cover all modifications and alternatives coming within the true scope and spirit of the invention as embodied in the attached claims. Applicant reserves the right to submit claims directed to combinations and sub-combinations of the disclosed inventions that are believed to be novel and non-obvious. Inventions embodied in other combinations and sub-combinations of features, functions, elements and/or properties may be claimed through amendment of those claims or presentation of new claims in the present application or in a related application. Such amended or new claims, whether they are directed to the same invention or a different invention and whether they are different, broader, narrower or equal in scope to the original claims, are to be considered within the subject matter of the inventions described herein.

Claims

1. A panel for a modular crate system, the panel comprising: a rectangular body comprising: an inner surface and an outer surface, wherein the outer surface is opposite the inner surface, and a

first edge, a second edge, a third edge, and a fourth edge, wherein the first edge is opposite the third edge, and wherein the second edge is opposite the fourth edge; and each edge of the first edge, the second edge, the third edge, and the fourth edge, comprises a plurality of teeth extending outwardly from the edge, wherein: each of the plurality of teeth are spaced apart such that the plurality of teeth is configured to intermesh with a plurality of teeth of an adjacent panel of the modular crate system to form at least a portion of a modular crate, each of the plurality of teeth includes a first tooth positioned at one end of the edge, the first tooth comprising a post configured to engage with a slot of a connector to couple the panel to the adjacent panel, and each of the plurality of teeth further comprises at least two different types of teeth, the two different types of teeth comprising: a first tooth type; and a second tooth type.

2. The panel of claim 1, wherein the first tooth type and the second tooth type are alternatively positioned along each of the plurality of teeth for at least a portion of a length of a corresponding edge.

3. The panel of claim 1, wherein each tooth of the first tooth type comprises an opening extending through the tooth from a first side face of the tooth to a second side face of the tooth.

4. The panel of claim 3, wherein each tooth of the second tooth type comprises an opening extending through the tooth from a third side face of the tooth to a fourth side face of the tooth.

5. The panel of claim 1, wherein between each tooth of the plurality of teeth: the inner surface of the rectangular body comprises a recess formed into the inner surface; and the outer surface of the rectangular body comprises a recess formed into the outer surface.

6. The panel of claim 5, wherein each recess of the inner surface and the outer surface is configured to receive a flange of a tooth of the plurality of teeth of the adjacent panel of the modular crate system.

7. The panel of claim 1, wherein the post comprises a cylindrical projection terminating in a circular flange.

8. The panel of claim 7, wherein the post comprises an opening extending along a longitudinal axis of the post.

9. The panel of claim 8, wherein the opening is threaded and configured to receive a bolt.

10. The panel of claim 9, wherein the bolt comprises an M8 bolt.

11. The panel of claim 1, wherein the rectangular body is square.

12. The panel of claim 11, wherein the panel is 40 inches by 40 inches by 1.5 inches.

13. A modular crate system, comprising: the panel of claim 1; and a connector for coupling the panel to another panel, the connector comprising: a cubic body comprising an upper surface and an opposed lower surface, and four side surfaces extending between the upper surface and the lower surface; an opening extending through the cubic body from the upper surface to the lower surface; and for each of the side surfaces, a T-shaped slot formed into the side surface and extending from the lower surface towards the upper surface.

14. The system of claim 13, wherein the opening of the connector is configured to receive a shaft of a bolt therethrough.

15. The system of claim 14, wherein the opening of the connector is not threaded.

16. The system of claim 14, wherein the bolt comprises an M8 bolt.

17. The system of claim 13, wherein the upper surface of the connector comprises a recess configured to receive a head of a bolt inserted through the opening.

18. The system of claim 13, wherein a closed end of each of the T-shaped slots is rounded.

19. The system of claim 13, wherein the cubic body of the connector is 1.5 inches by 1.5 inches by 1.5 inches.
